

FLUID STATICS

Two methods of analyzing fluids:

1. LaGrangian (follow a fluid element and describe its properties)
2. Eulerian (follow a point in space and look at its behavior over time)

Static fluids - no stress

Stress = force/area - it deforms fluids

Quantitative description of stress

Density $\rho = \frac{m}{v}$, Specific weight $\gamma = \frac{W}{V}$.

Specific gravity is similar to density, but it is relative to water: $SG = \rho/\rho_{water}$.

Bulk modulus $K = -\frac{\Delta p}{\Delta V/V}$ - measures the decrease in volume with increasing pressure. In other words, compressibility.

A force ($\vec{F}$) acting on a body has parallel and perpendicular components, and the surface has a normal $\vec{n}_A$.

Normal stress = $\lim_{A \rightarrow \delta A} \frac{F_{perp}}{A}$. (Basically, the infinitesimal derivative of force/area in a small area dA). Shear stress is the same for parallel force. Stress is a TENSOR, it is a three-dimensional vector (multiple direction). For the same system, τ_{xy}, τ_{yz} , etc... are not necessarily the same.

Pressure definition of stress $P = -\frac{1}{3}(\sigma_{xx} + \sigma_{yy} + \sigma_{zz})$. Positive in compression.

Pressure is a normal force that acts perpendicularly to a surface. Keep in mind the difference between absolute and gauge pressure: $P_{abs} = P_{atm} + P_{gauge}$.

Force balance basis: In each direction, there's a pressure gradient between a point x and a point $x + \Delta x$, and these pressures come against force gravity.

This takes the following form: $\rho \vec{g} - \vec{\nabla} P = 0$ - but pressure in a static fluid varies in the direction of gravity.

Think of the example of a u-tube manometer:

$\vec{\nabla} P = \rho \vec{g} = -\rho g \vec{e}_y$ (pressure gradient is dependent on density and direction of position relative to gravity)

So, $\frac{dP}{dy} = -\rho g$, $dP = \rho g dy$ (the pressure gradient is opposite to the direction of gravity)

So, the $P = \rho gh$ formula comes from integrating this.

Pascal principle: Points A and B at the same elevation have the same pressure if connected through the same fluid

FLUID DYNAMICS - CONSERVATION BASICS

Projection may be needed if not perfectly parallel/perpendicular to the unit normal.

Flux through a small area element: $\rho \vec{v} \cdot d\vec{A} = \rho v \cos(\theta) dA = \rho(\vec{v} \cdot \vec{n})$

Summation: $\int \int_S \rho(\vec{v} \cdot \vec{n}) dA = \text{efflux} - \text{influx}$

This equation takes the form of:

$$\frac{d}{dt} \int \int \int_{CV} \rho dV + \int \int_S \rho(\vec{v} \cdot \vec{n}) dA = 0$$

Relation of Newton's laws to momentum conservation: (here, p is momentum)

$$\frac{d}{dt}\vec{p} = \frac{d}{dt}(m\vec{v}) = m\frac{d}{dt}\vec{v} = m\vec{a} = \sum \vec{F}$$

$$\sum \vec{F} = \oint_{CV} \vec{v}\rho(\vec{v} \cdot \vec{n})dA + \frac{\partial}{\partial t} \iiint_{CV} \rho\vec{v}dV$$

Inlet:outlet problem force balance:

Force gravity in the downward direction, and then $P \cdot A$ for inlet and outlet.

S.S. - time term goes away, Incompressible: constant density, the equation becomes very simple

$$\nabla \cdot \vec{v} = 0.$$

Energy balances:

$$dE = \delta Q - \delta W, \frac{dE}{dt} = \frac{\delta Q}{dt} - \frac{\delta W}{dt} = \oint_{CS} e\rho(\vec{v} \cdot \vec{n})dA + \frac{\partial}{\partial t} \iiint_{CV} \rho e dV. \text{ (Same principle as mass balance)}$$

Work can be divided into shaft, countering shear, and countering normal stress. Normal work was moved over to the right side

$$\begin{aligned} \frac{\delta Q}{dt} - \frac{\delta W_s}{dt} - \frac{\delta W_\mu}{dt} &= \oint_{CS} e\rho(\vec{v} \cdot \vec{n})dA + \frac{\partial}{\partial t} \iiint_{CV} e\rho dV + \oint_{CS} P(\vec{v} \cdot \vec{n})dA \\ &= \oint_{CS} \left[e + \frac{P}{\rho} \right] \rho(\vec{v} \cdot \vec{n})dA + \frac{\partial}{\partial t} \iiint_{CV} e\rho dV \end{aligned}$$

e , which is energy per mass is the sum of kinetic, potential, and internal: $e = 0.5v^2 + gy + u$

This energy balance simplifies to the bernoulli eqn if the flow is steady, incompressible, inviscid, isothermal, no heat addition.

VISCOUS FLOW

Laminar flow: Smooth, no inertia, S.S possible, Turbulent has eddies, high velocity, no real S.S.

Viscous fluids: $\tau = c * \frac{d\gamma}{dt} = \mu * \dot{\gamma}$, where γ is deformation. Basically, shear stress is proportional to deformation rate, and viscosity relates the two.

$\tau(y) = \mu \frac{dv_x(y)}{dy}$ - Basically, the velocity gradient between adjacent fluid layers causes viscous stresses. Constant viscosity = newtonian

Viscosity - conceptual: Molecules move between adjacent fluid layers.

$$\mu = f(m, T, d) \text{ (temp, mass, size)}$$

Gases: μ increase w/ T, more movement between layers

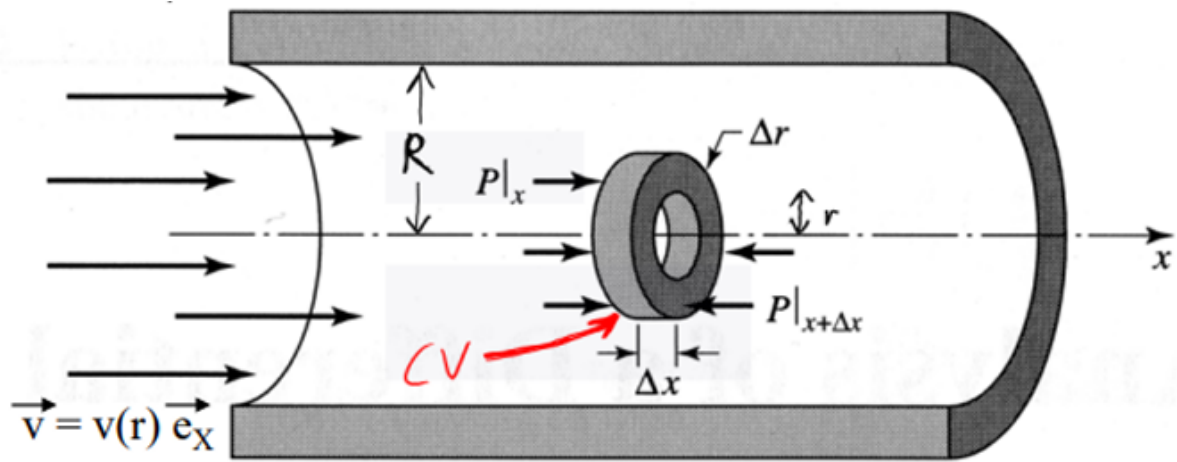
Liquids: Opposite, T increases intermolecular distance and decreases strength of IMFs.

Exercise: Derive the units of dynamic μ and kinematic ν viscosities. Hint: Look at the equation

Deformation can be in multiple directions: $\tau_{yx} = \tau_{xy} = \mu \left(\frac{dv_x}{dy} + \frac{dv_y}{dx} \right)$. This is where tensor notation comes in.

Exercise: Think closely about what value τ_{xx} takes and derive the stress tensor

$$\text{Force on surface A: } \vec{F} = \tau \vec{A} = \tau \cdot A \vec{n}.$$



Shell balance: take a small slice of Δx in your control volume, and balance force:

$$\Sigma F_x = P_{left} * 2\pi r dr - P_{right} * 2\pi r dr + \tau_{rear} * 2\pi r \Delta x - \tau_{front} * 2\pi \Delta x$$

(Basically, net force = net pressure force + net shear stress force)

$$\oint_{cS} \vec{v} \rho (\vec{v} \cdot \vec{n}) dA = \iint_{\text{entry area}} v_x \rho (-v_x) dA + \iint_{\text{exit area}} v_x \rho (+v_x) dA = 0$$

(net force is the sum of flow forces)

So, charting this out that this system:

$$-r \frac{dP}{dx} + \frac{d}{dr} (r \tau_{rx}) = 0$$

(pressure forces balance stress forces)

$$\frac{d}{dr} (r \tau_{rx}) = r \frac{dP}{dx}$$

Eventually, solve for τ .

$$\tau_{rx} = \frac{r}{2} \frac{dP}{dx}, \text{ with BC } \tau_{rx} \text{ finite at } r = 0$$

For a newtonian fluid: $\tau = \mu * \frac{dv_x}{dr}$, so you can solve for the velocity profile

Dimensional analysis

Existence of fundamental and derived (functions of fundamental) units. E.g. length (m) is fundamental, time (s) is fundamental, but velocity (m/s) is derived.

MLT (mass length time) comprises a units system, sufficient to describe a path.

As an example, in the MLT unit system is force ($\text{kg} \cdot \text{m} / \text{s}^2$), so MLT^{-2} .

To see if three quantities are dimensionally independent, put them in terms of your units system, make a matrix with the quantities as columns and your units (MLT) in the rows. If that matrix has linearly independent columns ($\det \neq 0$ for a square matrix) then the quantities are dimensionally independent.

Dimensionless groups are unitless compilations of dimensional quantities, and can inform about a lot of things. For instance, Re, which is inertial force/viscous force.

MASS AND MOMENTUM CONSERVATION

Shell balance interpretation of mass conservation ρv in and out (mass flow). In one direction, this looks like:

Net efflux (x) = $\rho v_x|_{x+\Delta x} \Delta y \Delta z - \rho v_x|_x \Delta y \Delta z$. Put it together to make the continuity equation:

$$\vec{\nabla} \cdot (\rho \vec{v}) + \frac{\partial \rho}{\partial t} = 0$$

In this case of constant ρ , $\vec{\nabla} \cdot \vec{v} = 0$. Simplify the continuity eqn. Without this assumption:

$$\begin{aligned}\vec{\nabla} \cdot (\rho \vec{v}) + \frac{\partial \rho}{\partial t} &= 0 \\ \rho \nabla \cdot \vec{v} + \vec{v} \cdot \vec{\nabla} \rho + \frac{\partial \rho}{\partial t} &= 0 \\ \rho \nabla \cdot \vec{v} + \frac{D}{Dt} \rho &\end{aligned}$$

$\frac{D}{Dt}$ is a substantial derivative. It's a more complete derivative. Conceptualize this with the total differential.

$$\begin{aligned}dT &= \frac{\partial T}{\partial t} dt + \frac{\partial T}{\partial x} dx + \frac{\partial T}{\partial y} dy + \frac{\partial T}{\partial z} dz \\ \frac{dT}{dt} &= \frac{\partial T}{\partial t} + \frac{\partial T}{\partial x} v_x + \frac{\partial T}{\partial y} v_y + \frac{\partial T}{\partial z} v_z = \frac{DT}{Dt}\end{aligned}$$

If we repeat this exercise for momentum, and balance shear stress: ($\rho \vec{v} v_x$ in the x-direction for instance), you get the Navier-stokes eqn:

$$\begin{aligned}\rho \frac{D\vec{v}}{Dt} &= \rho \vec{g} - \vec{\nabla} P + \mu \vec{\nabla}^2 \vec{v} \\ \frac{D\vec{v}}{Dt} &= \vec{g} - \frac{1}{\rho} \vec{\nabla} P + \nu \vec{\nabla}^2 \vec{v}\end{aligned}$$

In english: Remember that $F = ma$, and so the velocity profile depends on force balance. Fluid velocity depends on gravitational force, pressure force, and viscous force!

This is for laminar flow, incompressible, and constant viscosity (last two assumptions allow us to change divergence into laplacian). Of course, take away these assumptions, and the eqn gets more complex. For inviscid flow, this last term goes. This eqn is a common choice for solving for velocity profiles, and is the differential version of the earlier momentum eqn.

Simplifications that can eventually be done to this eqn:

1. Cancel the viscous term for inviscid flow
2. Steady state - no t dependence
3. Simplify the laplacians/gradients based on directionality of velocity.
 1. Radial symmetry, no angle dependence (different symmetries in different geometries)
 2. Fully developed flow = boundary layer is dominated, viscous forces dominate, there would be no velocity dependence on the direction of flow (z in a pipe) - basically, no dependence on position WITHIN a pipe
 3. Laminar - no eddies/vortices, so in a pipe, for instance, no radial flow

FLOW REGIMES

Flow regimes depend on the Reynolds number, more inertial to viscous forces --> more turbulent flow because the fluid elements have high momentum and can't be restricted to orderly motion.

$Re < 2300$ - Laminar, $Re > 23000$ - Turbulent, $2300 < Re < 23000$ - Transition regime

Drag coefficient is also a function of Re (and as such, a function of L, ρ, μ, v)

More detailed flow regimes:

1. $Re < 1$, viscous forces dominate, and the flow is completely stable - decreasing drag coeff with Re

2. Larger Re - viscous forces not dominant enough, eddies form - decreasing drag coeff with Re
3. Very large Re, increasing disorder at separation point, laminar flow in frontal region - stable drag coeff
4. Extremely large Re - turbulent flow every, quick drop in drag coefficient

Concept of a boundary layer:

Thin layer of fluid near a surface that is influenced by the surface rather than (or in addition to) the bulk. Turbulent boundary layer (which have higher friction) and laminar boundary layers differ in the flow pattern/presence of eddies. Essentially, the dictating factors for this layer are the no-slip condition at the surface.

Scaling analysis allows us to work towards understanding PDEs without actually solving. Basically, variables are put in terms of relevant scales, so for instance, scale $\frac{\partial}{\partial y} \sim \frac{1}{\delta}$, where δ is a dynamical length scale (we can apply this to the boundary layer, where the x length scale is just x, and the y length scale is δ)

We can apply this to Navier Stokes, where the $\frac{\partial^2 v_y}{\partial y^2} \sim \frac{v_\infty}{\delta^2}$. In the BL equations, we can focus on v_x NSE, as the $v_y \ll v_x$ due to the nature of the boundary layer. Think about it this way: $v_y = \frac{\delta}{x} v_\infty$, so negligible compared to x-direction because the length scale is much more than BL length scale. So $\frac{\partial P}{\partial x} = \frac{dP}{dx}$.

With these simplifications, as well as the steady-state assumption, the BL equations simplify to:

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0 : \text{Continuity eqn, or Reynolds Lubrication eqn}$$

$$v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} = \frac{\mu}{\rho} \frac{\partial^2 v_x}{\partial y^2} : \text{NSE for boundary layer}$$

B.C.:

At $y = 0$ (surf) : $v_x = 0 = v_y$ no slip at surface

At $y = \infty$ (bulk) : $v_x = v_\infty, v_y = 0$ convergence to bulk conditions away from surface

So, we can think of the velocity profiles as: v_x increasing from 0 to bulk as we get further from the surface, and v_y parabolically being 0 on both the bulk and surface ends.

Hard to solve, but we can scale the NSE in the y-direction to get important info:

$$\frac{\delta}{x} v_\infty \frac{v_\infty}{\delta} \sim \frac{\mu}{\rho} \frac{v_\infty}{\delta^2}$$

$$\delta \sim \sqrt{x}, \quad \delta \sim \frac{x}{\sqrt{Re_x}}$$

Blasius had correlations for these boundary layer thicknesses that depend on the flow regime.

PIPE FRICTION

Pipe flow: friction leads to head loss (energy loss that impacts Bernoulli balance)

$$h_L = \frac{\Delta P}{\rho g} = 2 f_f \frac{L}{D} \frac{v_{avg}^2}{g}$$

where f_f is the fanning friction factor. For laminar flow, $f_f = 16/Re$.

In some theoretical situations, f_f can be solved for analytically, but most often, Moody plots are used, one to find relative roughness and another that uses e/D and Re to solve for fanning factor.

Other areas of flow obstruction, which lead to friction: valves, expansions/contractions, bend/elbows. Head losses can be calculated for all of these things, and summed up to get the total head loss.

Modified bernoulli equation that sheds inviscid assumption:

$$\frac{\delta Q}{\delta t} - \frac{\delta W_s}{\delta t} = \dot{m} \left[\frac{V_{out}^2 - V_{in}^2}{2} + g(z_{out} - z_{in}) + \left(\frac{P_{out}}{\rho_{out}} - \frac{P_{in}}{\rho_{in}} \right) + gh_L \right]$$

Pumps:

Pump efficiency is the ratio of usable work to total work, so the power used by pressure head, static head, and velocity head over the total supplied power:

$$\eta = \frac{\dot{m}g * \Delta \left(\frac{P}{\rho g} + z + \frac{v^2}{2g} \right)}{\text{power supplied}}$$

Maximum height is restricted by head loss - use Bernoulli's equation for this. Built on the concept of NPSHA (Net pressure suction head available)

$$\frac{P_{rec}}{\rho g} - \frac{P_{vapor}}{\rho g} + (z_{res} - z_{in}) - h_{L,in}$$

Flow through packed bed:

Void fraction - fraction of available volume: $\epsilon = \frac{V_{voids}}{V_{voids} + V_{solids}}$. Also called porosity.

Superficial velocity - consider as if it was not going through a packed bed: $v' = \frac{Q}{A}$. Actual velocity depends on porosity: $v = \frac{Q}{\epsilon A}$.

Get an equivalent diameter using particle diameter, shape factor, and porosity, and then get a Reynolds number using this.

Find the pressure drop: $\Delta P = \rho g h_L$. Considering laminar and turbulent flow and friction together, we get the following equation (Ergun's equation)

$$\Delta P = \frac{150 \mu v' H}{(D_p \Phi_s)^2} \frac{(1 - \epsilon)^2}{\epsilon^3} + \frac{1.75 \rho (v')^2 H}{D_p \Phi_s} \frac{1 - \epsilon}{\epsilon^3}$$

Fluidized beds: at high superficial velocity, particles are lifted. Balance pressure and gravity forces for fluidization criteria:

$$\Delta P * A = V_{part}(\rho_{particle} - \rho_{fluid})g = A * L_{mf} * (1 - \epsilon_{mf}) * (\rho_{particle} - \rho_{fluid})g$$

Where L_{mf} is the height at minimum fluidization.

Heat transfer:

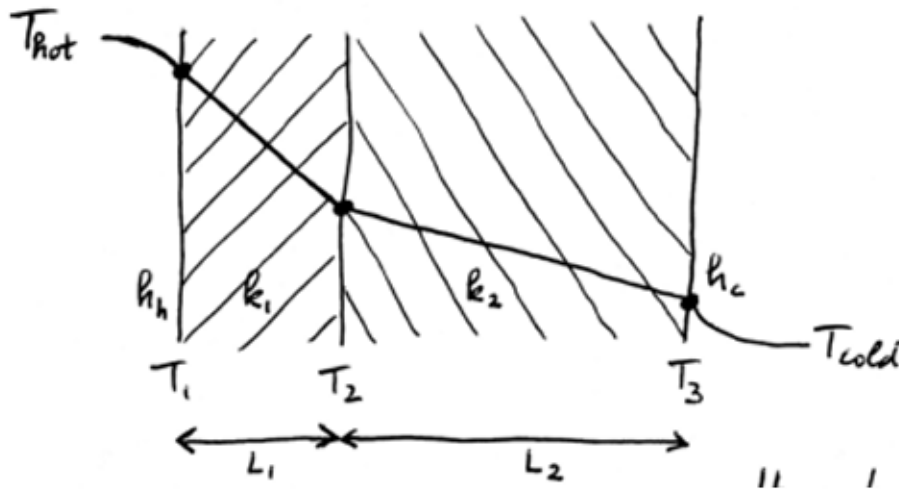
Three mechanisms: Conduction (from molecular collisions), convection (from fluid motion), radiation.

Heat flux $\vec{q}$ travelling in the direction of propagation - this concept leads to Fourier's law in one and multiple directions: $\frac{q_x}{A} = -k \frac{dT}{dx} (-k \vec{\nabla} T)$, (signage means that heat flows from high temp to low temp, as one would intuitively expect). k is the thermal conductivity. (Generally increases with T , but not always, higher for solids than for liquids/gases)

Newton's law of cooling (convection): $\frac{q}{A} = h \Delta T$. h is the convective heat transfer coefficient, represents the mass transfer in the boundary layer. h is higher for forced convection. **h depends on origin of flow (natural/forced), geometry of interface between surface and fluid, fluid properties.**

Radiation: Stefan-boltzman law: $\frac{q}{A} = \sigma T^4$, only significant at very high temps.

Combination of transfer mechanisms: take the below system



Convection from T_{hot} to T_1 : $q = h_1(T_{hot} - T_1)$

Conduction from T_1 to T_2 : (fourier) $q = k_1 A \frac{T_1 - T_2}{L_1}$ (equally valid is $k_1 A \frac{T_2 - T_1}{L_1}$)

Conduction from T_2 to T_3 : $q = k_2 A \frac{T_2 - T_3}{L_2}$

Convection from T_3 to T_{cold} : $q = h_2 A (T_2 - T_{cold})$

Heat flows are all equal, so solve for temp changes:

ex. $T_{hot} - T_1 = \frac{q}{h_1 A}$, $T_1 - T_2 = \frac{q L_1}{k_1 A}$, and so forth.

Series resistor analogy: Think of each transfer process as a resistor, q is current, ΔT total is voltage, and then those individual terms, the $\frac{1}{h_1 A}$, etc... are resistances.

Remember this:

$$q = \frac{(\Delta T)_{total}}{\Sigma R_{th}}$$

For composites, parallel resistor analogy will need to be used as well.

Multidimensional heat transfer:

$$\rho C_p \frac{DT}{Dt} = k \nabla^2 T + H_V (or \dot{q})$$

If k was constant, and we're in steady state, without heat generation, this is $\nabla^2 T = 0$.

$$\rho C_p V \frac{\partial T}{\partial t} = \dot{q} + k \nabla^2 T$$

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T + \frac{\dot{q}}{\rho C_p}, \text{ where } \alpha = \frac{k}{\rho C_p} \text{ (thermal diffusivity)}$$

T is some function $T(x,y)$, and to solve you can think of the Ansatz: $T(x,y) = X(x) \cdot Y(y)$ in steady state.

Transient heat transfer - same continuity equation with the transient term back in. Use PDE-solving techniques.

Simplified case where T is only dependent on time, start with initial condition $T(all \ r) = T_0$.

Energy balance when ignoring kinetic and potential energy:

$$\frac{\delta Q}{dt} = \frac{\partial}{\partial t} [\rho C_p T \int \int \int dV]$$

at constant volume: $\rho C_p V \frac{dT}{dt} = \rho C_p V \frac{\partial T}{\partial t}$

$q = hA(T_{\infty} - T) = \rho C_p V \frac{dT}{dt}$. Convective flux in a reservoir of temperature T_{∞} . This is actually analytically solvable using simple separability, and the final result, using the initial condition:

$$\theta = \frac{T - T_{\infty}}{T_0 - T_{\infty}} = \exp\left[-\frac{hA}{\rho V C_p} t\right]$$

(Concept of dimensionless temperature is very useful for heat transfer problems)

The exponential term can be divided into two parts:

$$\frac{h}{k} L_{char} * \left(\frac{\alpha t}{L_{char}^2}\right) = Bi * Fo$$

Bi is a dimensionless number (Biot) that gives the relative importances of conduction/convection in the system. Fo (fourier) gives an idea of the "speed of thermal conduction". This derivation applies when $Bi \ll 1$, so conduction outweighs convection. L_{char} is a characteristic length such that $L_{char} = V/A$.

What does this theoretical limit actually mean? Well, if conduction is much, much faster than convection, the BiFo is a **lumped parameter** that essentially is telling us that **functionally, the entire object is changing temp together, so $T = f(t)$ only.**

In the opposite case, which can be solved by FFT ($Bi \gg 1$, convection $\gg$ conduction) - assume a simpler system, a cartesian slab:

$$\frac{T - T_{\infty}}{T_0 - T_{\infty}} = \frac{4}{\pi} \sum_{n \text{ odd}} \frac{1}{n} \sin\left(\frac{n\pi}{L} x\right) \exp\left[-\left(\frac{n\pi}{2}\right)^2 Fo\right] = f(x, Fo)$$

The core temperature (hottest/coldest point) is then some function $g(Fo)$ such that (basically, sub $x = L/2$):

$$\frac{T_{center} - T_{\infty}}{T_0 - T_{\infty}} = \frac{4}{\pi} \sum_{n \text{ odd}} \frac{1}{n} \sin\left(\frac{n\pi}{2}\right) \exp\left[-\left(\frac{n\pi}{2}\right)^2 Fo\right] = f(x, Fo)$$

Charts exist here, this can be hard to solve analytically (Heissler charts). What can we say about a theoretical limit where convection dominates conduction? It would mean that heat transfer is dictated by the outside, $T_{surf} = T_{\infty}$.

Without either theoretical limit (also uses heissler charts) $Bi \sim 1$ (in terms of order of magnitude). In more advanced problems where it's not so easy to solve for Bi, start with this assumption. There is where you get the typical B.C.:

$$-k \nabla T|_{surf} = h(T_{surf} - T_{\infty})$$

This B.C., unlike the other two, accounts for the combination of conduction and convection. What does this really mean? We are now accounting for the existence of a heat transfer . (A: Boundary layer)

A finite body can be thought of as the intersection of multiple infinite bodies. So, for instance, a finite cylinder is a combination of an infinite slab and infinite cylinder. What infinite shapes would need to be combined to get a cartesian rectangular prism? (A: Two infinite slabs)

More on radiation:

One important property here - remember Stefan-Boltzmann law: $\frac{q}{A} = \sigma T^4$. How is this different from other heat transfer equations? (A: depends on T itself, not T difference)

Radiation can either be transmitted, reflected, or absorbed: $\tau + \rho + \alpha = 1$. (transmissivity, reflectivity, absorptivity). Ideal absorption is when $\tau, \rho = 0$, $\alpha = 1$.

Radiation emitted from a surface can be thought of as a flat surface into a hemisphere because it is emitted diffusely in all directions where it is visible:

$$\frac{dq}{dA} = E = I \int \cos\theta d\Omega$$

In words, emissive power is dependent on radiation intensity and the geometry (specifically, angles)

Stefan-boltzmann is ideal, so it calculates emission from a black body. For real surfaces: $E = \epsilon E_b$, or emissive power is the black body's emissive power times an emissivity. Between two bodies, to find q you look at the difference between emissivities $E_1 - E_2 = \epsilon\sigma(T_1^4 - T_2^4)$, and then an area integral. I and θ come from the emitting body. so you would end up essentially subtracting emission from 1 --> 2 from 2 --> 1.

END TRANSPORT 1

Ratio of momentum to heat diffusivities (from NSE and energy balance) can be expressed as the prandtl number:

$$Pr = \frac{\mu C_p}{k}$$

This isn't a flow property, it's a material property. Any given material has a tendency towards a certain level of viscous/thermal diffusion.

Boundary layer in heat transfer represents the fact that T, like v transitions from surface to bulk.

So, we equate conduction and convection:

$$\begin{aligned} q_{cond} &= q_{conv} \\ h(T_s - T_\infty) &= -k \vec{\nabla} T \\ h(T_s - T_\infty) &= -k \vec{\nabla} T = k \frac{d(T_s - T_\infty)}{dy} \Big|_{y=0} \end{aligned}$$

Q: Put yourself through an exercise while reading this - rationalize to yourself why the y direction is the chosen one for the derivative (assuming the surface is oriented in the x direction). Then explain the directionality. Why did the sign of k change? Why is the boundary condition at $y = 0$?

A: The y-direction is the direction of δ , the boundary layer thickness. It would be going perpendicular from the surface, originating AT the surface (hence $y = 0$), going up towards the bulk. The derivative is showing the instantaneous derivative of surface to bulk (really it's $T - T_\infty$ but at the surface, $T = T_s$). The $T_s - T_\infty$ goes against the gradient of temperature, as that would be in the direction of decreasing y, so that's why the sign flips.

Further manipulations lead us to the Nusselt number $Nu = \frac{hL}{k}$, which gives us the ratio of convection to conduction in a fluid, but especially in the boundary layer.

Dimensional analysis tells us how h relates to certain quantities. For forced convection, you worry about the geometry, flow, fluid properties, and for natural, you add the element of caring about the cause of flow (as it's natural)

You can start seeing ways in which dimensionless numbers relate:

Forced : $Nu = \phi(Re, Pr)$ or $St(Stanton) = \frac{h}{\rho v C_p} = \phi(Re, Pr)$ depending on chosen dimensional core

Natural : $Nu = \phi(Gr, Pr)$

$$Gr(Grashof) = \frac{\rho^2 L^3 g \Delta T \beta}{\mu^2} = \frac{\text{bouyancy forces}}{\text{viscous forces}}$$

Now that we've started to form an edifice for the boundary layer in heat transfer, form boundary layer equations:

$$v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2}$$

B. C. $T(y=0) = T_s$, $T(y \rightarrow \infty) = T_\infty$

Blasius correlations:

BL thickness: $\delta = 5\sqrt{\frac{\nu_x}{v_\infty}}$, $\frac{\delta}{x} = \frac{5}{\sqrt{Re_x}}$. Skin friction: $C_f = 0.664Re_x^{-0.5}$ (relations slightly changed based on regime)

$Nu \propto Re, Pr$. Reynolds analogy: $St = 0.5C_f$. Colburn: $0.5C_f = StPr^{2/3}$. Prandtl: $St \propto C_f, Pr$.

Don't worry too much abt the specific correlations, understand the relations.

Phase changes:

Boiling:

$$\frac{q}{A} = hT_e$$

Heat flow in boiling is driven by excess temp ($T_e = T - T_{sat}$). Different regimes exist here:

1. $T_e < 5$ - No boiling, just free convection
2. $5 < T_e < 10$ - Initial nucleation of bubbles
3. $10 < T_e < 30$ - Maximum q/A , boiling with bubbles
4. $30 < T_e < 120$ - Decreased heat flux, unstable vapor film insulates hot surface
5. $T_e > 120$ - Stable vapor film between solid/liquid, conductive heat transfer. If it's really high, radiation comes into play.

Correlations exist for the boiling heat, don't worry too much about them.

For condensation, correlations exist for h - the most important thing here is films/BLs form during phase change.

Heat exchangers: Can be counterflow or co-flow - there's an inner tube fluid and an outer tube fluid that is

$q = UAF\Delta T_M$. A = area, ΔT_M is a log mean temperature difference, and U is an overall heat transfer coefficient. F is a correction factor for more complicated design of heat exchanger - for simple single-pass shell-and-tube: $q = UA\Delta T_M$.

You can use the overall heat transfer resistance to calculate U :

$$UA = \frac{1}{\Sigma R_{th}} = \left[\frac{1}{h_{inner}A} + \frac{1}{h_{outer}A} + \frac{\ln(R_o/R_i)}{2\pi kL} + R_{foul} \right]$$

Resistances are as follows:

1. Inner and outer fluid convective heat transfer resistance
2. Third term is specifically for thick tube walls that have their own resistance, depends on tube material (conductivity) and thickness.
3. Fouling resistance, optional but necessary for certain systems, especially older heat exchangers.

In addition, temperature change design equations are useful - often to equate Q 's:

$$q = \dot{m}_{hot} C_{p,hot} (T_{hot,in} - T_{hot,out})$$

$$q = \dot{m}_{out} C_{p,hot} (T_{cold,out} - T_{hot,in})$$

Basically, can equate your heat flows to the thermal energy change due to temp change for cold and hot. In the direction of increasing temp - keep in mind that the cold fluid gets hotter and the hot fluid gets colder.

When you don't know the outer temperatures:

$$q = \epsilon(\dot{m}C_p)_{\min}(T_{hot,in} - T_{cold,in})$$

The min term - whichever $\dot{m}C_p$ is lower between the hot and cold fluid. ϵ is the effectiveness of a heat exchanger. To get this, you use that minimum heat capacity, and figure out the number of transfer units: $NTU = \frac{UA}{(\dot{m}C_p)_{\min}}$ and then a graphical solution can be used.

MASS TRANSFER

Molecular transport is driven by concentration gradients (in actuality, it's chemical potential). Molecular collisions help heat conduction but hinder mass transfer, so opposite of heat - diffusivity is higher for gases than liquids.

Most common form of fick's law:

$$N_A = -CD_{AB}\nabla y_A + y_A(N_A + N_B)$$

Existing flux balance and concentration gradient dictate the flux.

Auto-diffusion - in gases, through itself. How can we measure this in an experiment? Isotopes!

$$D_{AA} = \frac{1}{3}\lambda u$$

Dependence on mean free path and mean velocity (How much space is there to move and how fast does it move?)

$$D_{AA} = \frac{1}{3} \frac{k_B T}{\sqrt{2}\pi\sigma_A^2 P} \sqrt{\frac{8N_A k_B T}{\pi M_A}} \sim \frac{T^{3/2}}{\sigma_A^2 P \sqrt{M_A}}$$

Filling in the terms does enlighten us as to the effects of collision diameter, pressure, temperature, molar mass.

Intermolecular interactions - **Lennard-Jones**

$$\Phi(r) = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} + \left(\frac{\sigma}{r}\right)^6\right]$$

ϵ is an energy of interaction strength, σ is range of interaction (molecular size), r is the center-center distance.

1. Very low r : Repulsive forces dominate
2. Middle r : Attractive forces dominate
3. High r : Attractive > Repulsive, but both die down

Hirschfelder equation:

$$D_{AB} = \frac{0.0018583}{P(\sigma_{AB})^2 \Omega_{AB}} T^{3/2} \sqrt{\frac{1}{M_A} + \frac{1}{M_B}}$$

This accounts for molecular properties but also intermolecular forces. Certain scaling relations are important here:

$D \propto T^{3/2}, \sqrt{M_A}, P^{-1}, \sigma^{-2}, \Omega^{-1}$. Alternatively, another form can be used that depends on diffusion molar volumes. (Exercise, what is the scaling of T, P, V, M in this equation?)

For diffusivity in liquids, a general form exists: $\frac{D_{AB}\mu_B}{T} = f(\text{molar volume})$

Stokes-einstein: $D_{AB} = \frac{k_B T}{6\pi R_A \mu_B}$ - for colloidal systems (Spherical, low Re)

Common eqn: Wilke-chang

$$\frac{D_{AB}\mu_B}{T} = \frac{7.4 * 10^{-8} (\Phi_B M_B)^{0.5}}{V_A^{0.6}}$$

Φ is an association factor, higher for more polar compounds. Higher with solvent molar mass (solvation effects), with more molar volume.

Diffusion regimes:

1. Knudsen - Mean free path is higher than pore radius - Wall collisions dominate $\lambda > 10r_p$ ($D \propto T^{0.5}$)
2. Bulk - Molecular collisions dominate $r_p > 10\lambda$ ($D \propto T^{1.5}$)
3. Transition - Intersection between the two.

In porous structure, diffusivity depends on porosity and tortuosity: $D_{eff} = D \frac{\epsilon}{\tau}$.

For transition regime:

$$\frac{1}{D_{eff,overall}} = \frac{1}{D_{eff,K}} + \frac{1}{D_{eff,bulk}}$$

Balancing flux in a control volume:

$$\vec{\nabla} \cdot \vec{N}_A + \frac{\partial c_A}{\partial t} - R_A = 0$$

Flux divergence, transient term, reaction rate balance with each other.

Convenient PDE form:

$$\frac{\partial C_A}{\partial t} + u \cdot \nabla C_A = D \nabla^2 C_A + R_A$$

Transient + convection = Diffusion + Reaction

Steady state diffusion: $N_B = 0$ (stagnant). $\vec{\nabla} \cdot \vec{N}_A + \cancel{\frac{\partial c_A}{\partial t}} - \cancel{R_A} = 0$, $\frac{dN_A}{dz} = 0$

So in fick's law, $N_A = -CD_{AB} \frac{dy_A}{dz} + y_A(N_A + \cancel{N_B})$ - this can be used to generate a flux expression.

Equimolar counterdiffusion, $N_A = -N_B$, so $N_A = -CD_{AB} \frac{dy_A}{dz} = D_{AB} \frac{C_{A1} - C_{A2}}{z_2 - z_1}$. leads to linear concentration profile

Mass transfer with reaction: Same solving methods with a reaction term. You're going to end up having to use PDE solving techniques. Let's practice with 1D steady state, first order:

$$\cancel{\frac{\partial C_A}{\partial t}} + \cancel{u \cdot \nabla C_A} = D \nabla^2 C_A + R_A$$
$$D \frac{d^2 C_A}{dz^2} = k C_A, \quad D \frac{d^2 C_A}{dz^2} - k C_A = 0$$

Possible B.C.

1. Dirichlet: $C_A(z = 0) = C_{A0}$
2. Neumann (no flux): $\frac{dC_A}{dz} \big|_{\text{boundary}} = 0$.

Transient mass transfer - add in transient term. Based on the medium, geometry, and initial/BC, you can have different fickian solution. For no convection, no reaction, it looks like this:

$$\frac{dC_A}{dt} = D \frac{d^2 C_A}{dz^2}$$

Diffusivities are there for all types of transfer: Momentum - $\nu = \frac{\mu}{\rho}$, Mass - D_{AB} , Thermal - $\alpha = \frac{k}{\rho C_p}$.

Lewis number: $Le = \frac{\text{thermal diffusion}}{\text{mass diffusion}} = \frac{\alpha}{D_{AB}} = \frac{Sc}{Pr}$.

Convective M.T.: $N_A = k_c \Delta C_A = k_c (C_{As} - C_{A,\infty})$. k_c is a convective mass transfer coefficient. Liken this to h for heat transfer.

Mass equivalent of nusselt number is Sherwood number (convection/conduction): $Sh = \frac{k_c L}{D_{AB}}$

Boundary layer analysis: Laminar flow across a flat plate: Correlation between Sh, Re, Sc:

$$Sh = \frac{kL}{D} = 0.332 Re^{0.5} Sc^{1/3}$$

Conduction/convection ratio (use this correlation to calculate a k_c) is a function of inertial forces, viscous forces, viscous diffusion, and mass diffusion

Convective mass transfer coefficient models:

Film theory for steady state convective mass transfer: Imagine as purely molecular diffusion over an film of thickness δ chosen to fit the proper thermal resistance. Eqn: $k_c \frac{D_{AB}}{\delta}$ (δ can be found at low Sc numbers)

Penetration theory: There is a certain level of penetration in a semi-infinite medium, convective transfer depends on velocity, penetration depth, diffusivity. Key question: What is the exposure time of some fluid parcel to the interface? Eqn is: $k_c = \sqrt{\frac{D_{AB} v_{\infty}}{\pi \delta}}$. Good for high Sc

BL theory: If one phase is solid, there's no relative motion at the interface, so we employ BL theory to account for surface and bulk effects (in line with Blasius result): $k_c = 0.664 \frac{D_{AB}}{L} (Re)^{0.5} Sc^{1/3}$.

Mass, heat and momentum transfer can't always be thought of separately, often they occur simultaneously.

When mass moves, energy is transported with it:

$$\frac{\vec{q}_D}{A} = \sum_i \vec{N}_i \bar{H}_i$$

In english: Heat flux from MASS diffusion = molar flux*partial molar enthalpy.

Consider the flux balancing we did in 6260 - let's take one direction, z in a stagnant film where A is moving and B isn't.

$$\begin{aligned} \vec{\nabla} N_A &= \frac{dN_A}{dt} (S.S.) = 0 \\ N_A &= -c D_{AB} \frac{dy_A}{dz} + y_A (N_A + \cancel{N_B}) \\ N_A &= -\frac{c D_{AB}}{1 - y_A} \frac{dy_A}{dz} \end{aligned}$$

However, in reality, in the gas phase, D, c are functions of T.

D scaling with T:

$$\begin{aligned} D &\propto T^{1.5} \text{ (Bulk)} \\ D &\propto T^{0.5} \text{ (Knudsen)} \end{aligned}$$

Interphase mass transfer:

Convective fluxes have different forms based on the stage;

$$N_A = k_L \Delta C_A$$

$$N_A = k_G \Delta P_A$$

$$N_A = k_y \Delta y_A$$

The deltas are concentration differences among a phase, but in interphase mass transfer, you're mixing driving force variables.

Two film theory bridges this gap with two simplifying assumptions:

1. No interfacial resistance to mass transfer
2. There are equilibrium relations between stages, e.g. $P_{A,i} = f(C_{A,i})$.

Raoult's law is an equilibrium relation

$$P_A = y_A P = x_A P_A^*$$

Self-test: what law allows us to transform the first to second form of Raoult's law above (A: Dalton's law)

So his Henry's law: $P_A = H x_A$ - for dilute solutions

At an interface you can equate fluxes - e.g. liquid-gas interface

$$k_G(P_{A,G} - P_{A,i}) = N_A = k_L(C_{A,i} - C_{A,L})$$

You can relate local mass transfer coefficients with driving forces:

$$-\frac{k_L}{k_G} = \frac{P_{A,G} - P_{A,i}}{C_{A,L} - C_{A,i}}$$

Overall MTCs involve equilibrium/saturation C and P, C^* , P^* that are hypothetical concs/pressures in equilibrium with the other phases.

$$K_G(P_{A,G} - P_A^*) = K_L(C_A^* - C_{A,L})$$

If there's a simple linear equilibrium relation $P_{Ai} = m C_{Ai}$, you can re-jig that above equation in terms of just one MTC

$$k_G(P_{A,G} - P_{Ai}) = k_G m (C_A^* - C_{Ai}) = N_A$$

You can then relate m:

$$m = \frac{C_A^* - C_{Ai}}{P_{A,G} - P_{Ai}}$$

So, you can then form relations between overall and local MTC:

$$\frac{1}{K_L} = \frac{1}{k_L} + \frac{1}{k_G m}$$
$$\frac{1}{K_G} = \frac{1}{k_G} + \frac{m}{k_L}$$

If $\frac{1}{k_G} = \frac{1}{K_G}$, gas phase controls and vice versa.

Interfacial mass transfer depends on contact area and time, so equipment will be made to maximize contact (rationality behind countercurrent designs and continuous contact equipment)